



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 4 • STANDARD 6.AT.4

Percents Greater Than 100% and Less Than 1%

Lesson 4-3 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can write and interpret percents greater than 100% and less than 1%.

Language Objective: I can explain my reasoning using the words percent, greater than 100%, less than 1%, and equivalent.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Percent

Greater than 100%

Less than 1%

Equivalent

Mixed number



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A number out of 100, written with a % sign, is a .

2

A percent that represents more than one whole, like 150%, is

.

3

A percent smaller than one out of a hundred, like 0.5%, is

.

4

Values that name the same amount, like 0.5% and 0.005, are

.

5

A number with a whole part and a fraction part, like $1\frac{1}{2}$, is a

.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

COMPARE

When you wrote percents like 150% and 0.5% as decimals, what did you notice about their size compared to 1?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Percent** — A way to compare a number to 100, shown with the % sign.
Porcentaje — Una manera de comparar un número con 100, con el signo %.

☐ **Greater than 100%** — A percent bigger than 100%, more than the whole thing.
Mayor que 100% — Un porcentaje mayor que 100%, más que todo el entero.

☐ **Less than 1%** — A percent smaller than 1%, a very tiny part.
Menor que 1% — Un porcentaje menor que 1%, una parte muy pequeña.

☐ **Equivalent** — Having the same value, just written a different way.
Equivalente — Tener el mismo valor, escrito de otra forma.

☐ **Mixed number** — A whole number plus a fraction, like $1 \frac{1}{2}$.
Número mixto — Un número entero más una fracción, como $1 \frac{1}{2}$.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

150% as a decimal is ____ because ____.

150% como decimal es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: 150% is more than one whole, and 0.5% is a tiny piece smaller than 1%.

Evidence: Students explain 150% is more than one whole (the player beat the goal) and 0.5% is far less than one whole (a tiny boost), connecting to the idea of 100% as the whole.

Reasoning: This evidence connects the definition of percent to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **150% as a decimal is ____ because ____.**
150% como decimal es ____ porque ____.
- **Percents over 100% have decimals ____ than 1, and percents under 1% have decimals ____.**
Los porcentajes mayores que 100% tienen decimales ____ que 1, y los menores que 1% tienen decimales ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**
Lo sé porque ____.
- **So ____.**
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Percent
2. **Notes 2:** Greater than 100%
3. **Notes 3:** Less than 1%
4. **Notes 4:** Equivalent
5. **Notes 5:** Mixed number
6. **Watch for this mistake:** *A common mistake in Percents Greater Than 100% and Less Than 1% is treating every percent-to-decimal conversion the same way regardless of size — for example, writing 150% as 0.15 (as if it were an ordinary under-100% percent) instead of 1.5, or writing 0.5% as 0.5 (forgetting to divide by 100 at all) instead of 0.005. Before finalizing an answer, check that the decimal for any percent over 100% is greater than 1, and the decimal for any percent under 1% is less than 0.01.*
7. **Try It 1:** A. 0.005 — $0.5\% = 0.5 \div 100 = 0.005$. Because it is less than 1%, the decimal must be less than 0.01.
8. **Try It 2:** A. About 20 mAh — a tiny sliver, almost empty — $0.5\% = 0.005$, and $0.005 \times 4,000 = 20$ mAh. A percent less than 1% is a tiny part, so the phone is nearly dead.